

Retrofitting a commercial RF induction generator into a computer-controlled, vacuum and gas integrated annealing system for reactive-metal grain growth

Sterling G. Baird, Ryan Weber, Christopher Nyborg, Ronnie Guymon, Gage Erickson, and Oliver Johnson
Department of Mechanical Engineering, Brigham Young University, Provo, Utah 84602, USA

(Dated: 11 August 2026)

High-temperature vacuum annealing near a metal’s melting point drives controlled grain growth, but it normally requires an expensive turn-key vacuum induction furnace. We retrofit a bare commercial radiofrequency (RF) induction generator with computer power control through LabVIEW, dual-wavelength optical temperature feedback, and a high-vacuum quartz-tube chamber with inert-gas backfill. A machined graphite crucible doubles as a susceptor, an RF-absorbing element that heats the specimen it encloses. The only requirement is a monotonic analog power-control input, so the retrofit transfers between generators. Our reference build uses a 6 kW solid-state generator. Surprisingly, nickel annealed in this system produced high-quality electron backscatter diffraction (EBSD) patterns with zero specimen preparation. Samples went directly from the furnace to the electron microscope without grinding, polishing, or etching. The EBSD maps indexed hundreds of equiaxed grains delineated by deep thermal grooves. A modified sample assembly extends the furnace to ceramics that do not couple to the RF field. It coarsened yttria-stabilized zirconia (YSZ) grains in 45 min at 2500 °C, compared with 228 h at 1600 °C in a conventional box furnace. For a fixed configuration, the power–temperature calibration is linear ($R^2 = 0.991$) from 1200 to 1400 °C. Eight 12 h nickel anneals at 1200 °C reproduced their soak temperature to 1201.2 ± 1.3 °C, and 40 h soaks at 1325 °C remained stable. Complete design files, a bill of materials, control software, and data are openly available. This gives laboratories an affordable route to near-melting-point annealing with direct anneal-to-EBSD characterization.

Keywords: induction furnace; grain growth; vacuum annealing; LabVIEW; pyrometer; open-source hardware

I. INTRODUCTION

High-temperature annealing near a metal’s melting point drives controlled grain growth. This is central to studying microstructure–property relationships in metals such as nickel and iron^{1–5}. Growing grains without oxidizing the sample requires temperatures of 1400 to 1500 °C along with high vacuum or an inert atmosphere^{6,7}. Commercial vacuum induction furnaces that meet these requirements are closed, turn-key systems costing \$50k to \$200k or more. Resistive tube and box furnaces do not heat the charge directly. They heat primarily through radiation from the furnace elements, so heat transfer into the charge is limited, and they reach lower temperatures with slower ramps. Bare industrial induction generators are widely available, but they ship without vacuum, optical temperature feedback, or computer setpoint control.

This work adds those missing subsystems as an open, documented retrofit. A bare RF induction generator^{8,9} is retrofitted into a full annealing system with computer power control, closed-loop optical temperature feedback, a high-vacuum quartz-tube chamber, and a crucible that doubles as a susceptor. The retrofit is deliberately generator-agnostic. Its only requirement is an analog power-control input with a monotonic command-to-power response. The pyrometer feedback and proportional–integral–derivative (PID) calibration absorb any nonlinearity in that response. The generator internals are not reproduced, so a compatible generator needs to be supplied. A graphite crucible couples

strongly to the RF field and is used for the metal grain-growth anneals. Non-coupling ceramics such as YSZ use a modified sample assembly (the specimen, susceptor, and supports inside the chamber) in which graphite or tantalum serves as the susceptor, chosen for chemical compatibility with the specimen (Sec. V A).

The reproducible build documented here is anchored on a modern 6 kW solid-state generator whose controller accepts a standard analog power setpoint. The reference generator is a CEIA “Power Cube” PW3-90/50 with its matching controller, heating head, and chiller. The generator package and the retrofit components are itemized in the bill of materials in the supplementary material. The cost of the entire system comes to about \$38k.

II. SYSTEM DESIGN AND DESCRIPTION

The system comprises five subsystems: heating, vacuum and gas handling, optical temperature sensing, computer control, and mechanical support (Fig. 1(a)). Figure 1(b) shows the assembled system at operating power.

The heating subsystem is a 6 kW solid-state induction generator driving a liquid-cooled copper work coil^{10,11}. In our case, we procured a coil from CEIA that is approximately 3 in tall with a 2.5 in inner diameter and 6.5 turns, tailored to our vacuum chamber geometry. The coil conductors carry both the RF current and the coolant. Ethylene glycol coolant flows in series from the chiller through the generator and then through the heating head and work coil before returning to the chiller.

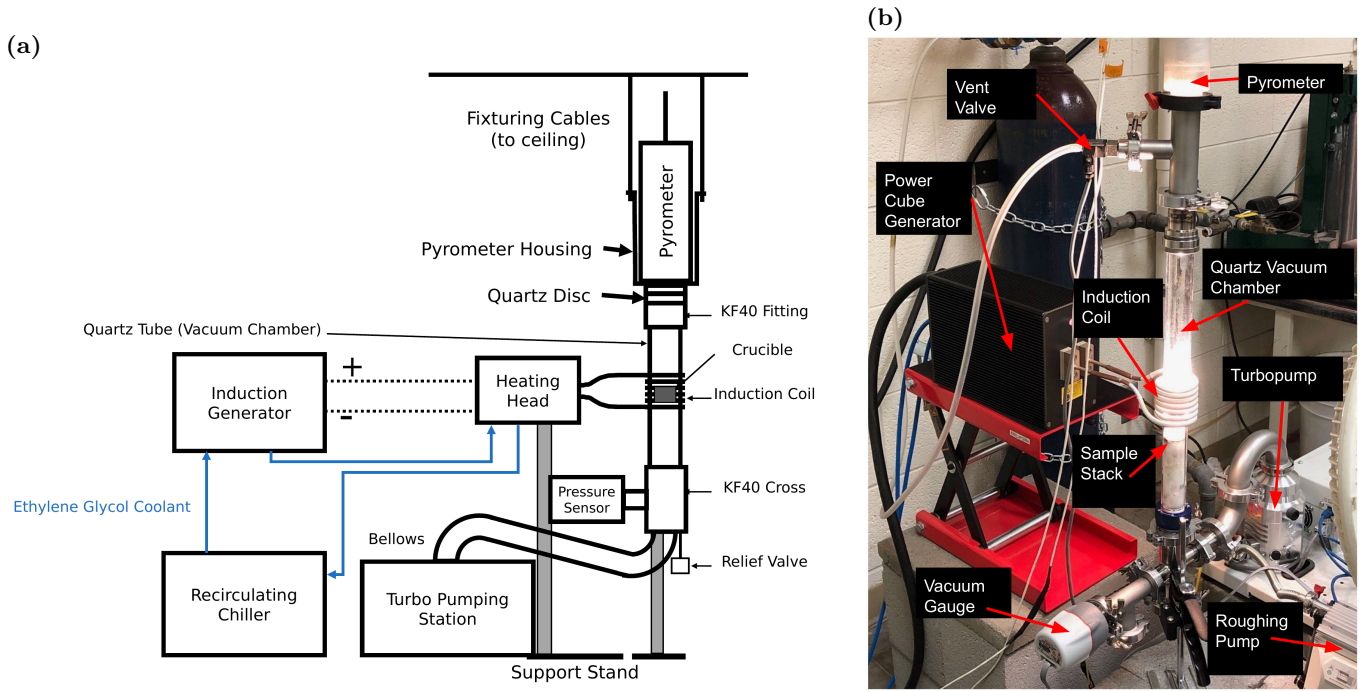


FIG. 1. The annealing system. (a) System overview. The generator drives the liquid-cooled heating head and work coil around the vertically suspended quartz-tube chamber. The quartz tube is joined by KF40 flanges to the ceiling-suspended pyrometer housing above and to the vacuum hardware below. A smaller support stand under the vacuum chamber supports the vacuum hardware from below. An overpressure relief valve and a flexible bellows lead past the pressure sensor to the turbopumping station. (b) The assembled, functioning furnace with the main components labeled. This shows the quartz chamber glowing at the sample-assembly position. The pyrometer sights down the chamber axis, and the vent valve admits the argon backfill. The gauge, turbopump, and separately supported roughing pump complete the vacuum path. Relative to closed turn-key vacuum induction furnaces, this design costs far less, is modular (hand-clamped KF-flanged vacuum hardware and an interchangeable sample assembly), transparent (customizable LabVIEW control software and design files), and it reaches a high-purity annealing environment by pumping the chamber to high vacuum and then backfilling with flowing argon.

The vacuum and gas subsystem starts from the 35 mm inner-diameter fused-quartz tube that forms the chamber. The tube connects through a KF40 flange (a standard clamp-sealed vacuum flange) and a flexible bellows to a compact turbopumping station that reaches 10^{-6} to 10^{-8} Torr. An automated, electrically triggered purge vent valve fed from an inert-gas regulator allows controlled venting and backfilling. These components support a range of operating atmospheres. The furnace can run at low or high vacuum, and the chamber can instead be backfilled with inert gas and continuously supplied either at a precisely controlled rate, for example 20 standard cubic centimeters per minute (SCCM), or as a large purge flow. The turbopump can be left running during backfilled operation or stopped. The atmosphere matters for graphite-crucible anneals^{12,13}. The best results were achieved by pumping to high vacuum and then backfilling with argon through the purge vent valve. The chamber was purged first, and a constant argon flow of 20 SCCM was then maintained for the duration of the anneal using a mass flow controller. Many of the reported nickel grain-growth anneals ran under this continuous flow. Heating under vacuum alone tended to sublime graphite onto the chamber walls. The highest-temperature ceramic work

(Sec. V A) used the same continuous argon supply, with the chamber evacuated and then filled with argon while the turbopump was running. This helped avoid oxidation during heating.

Temperature is sensed optically. A dual-wavelength ratio pyrometer^{14,15} with an 800 to 2500 °C sensing range views the sample through a quartz optical window. This provides a non-contact signal that is robust to emissivity changes and to the RF field that would corrupt thermocouples^{16–18}. Control is implemented in LabVIEW, with virtual instruments (VIs) for manual control, automated ramping and PID tuning. The generator's power setpoint can be commanded either by a 0–5 V voltage signal or by a 4–20 mA current-loop signal. Which input is available depends on the generator model and its configuration. In this build, the data-acquisition (DAQ) device's 0–5 V analog output drives a voltage-to-current loop conditioner, and the conditioner's 4–20 mA output drives the generator's power-setpoint input. The pyrometer's analog output returns to a DAQ input, which closes the PID loop. Below the pyrometer's 800 °C detection floor, the VI ramps open-loop. Once the pyrometer returns a sustained valid reading, the VI switches to closed-loop PID control based on the optically-measured tempera-

ture^{19,20}. Mechanically, a bolted support stand carries the generator's heating head and centers the work coil on the crucible position. The quartz tube and the vacuum hardware below it are joined by KF40 flanges to the pyrometer housing, which is suspended from the ceiling by cables. A second, smaller support stand sits under the vacuum chamber and supports the vacuum hardware from below. The sample assembly is not attached to any of these flanges. It simply sits inside the quartz tube, resting on a retaining ring of a KF40 flange.

Metal charges in the graphite crucible reach the nickel and iron grain-growth range of 1400 to 1500 °C, with demonstrated soaks (constant-temperature holds) from minutes to 40 h. The modified ceramic sample assembly extends the working range to 2500 °C. When the retrofit is moved to a different generator, a user would need to re-determine optimal parameters specific to their system; however, the general framework is agnostic to the generator type and brand.

A. Graphite crucible / susceptor

Metal specimens are held in a two-piece graphite crucible machined in-house from 3000 °C-grade graphite stock. It consists of a cup body and a lid with a central bore, which contains the charge while leaving an optical path for the pyrometer. Figure 2(a) shows the disassembled parts. The graphite couples strongly to the induction field and acts as a susceptor, heating the contained metal specimen by conduction and radiation. The specimen is sandwiched between two alumina discs in the cup, the lid sits on top, and a sapphire window closes the lid bore (Fig. 2(b-d)). Sapphire is used for the window to increase transmittance in the wavelength ranges emitted from the sample relative to the pyrometer and because it has a higher melting point than quartz. The measured dimensions of every piece are given in the supplementary material. Inside the vacuum chamber the crucible rests on an alumina support tube that positions it at coil height. The tube has holes bored laterally along its outer perimeter to aid evacuation, and its lower end rests on top of a Teflon tube at the KF40 fitting below. The crucible is used primarily for the metal grain-growth anneals. Graphite also works as a susceptor for some ceramic charges when the contact chemistry permits. Sec. V A describes how a compatible susceptor is selected for refractory ceramics.

III. CONSTRUCTION AND OPERATION

Following the heating path, the generator's heating head mounts on an adjustable platform on the bolted support stand so the coil centers on the crucible position of the vertically suspended quartz tube. The vacuum path begins at the both-ends-open quartz tube. Its lower end joins the vacuum hardware below through a KF40

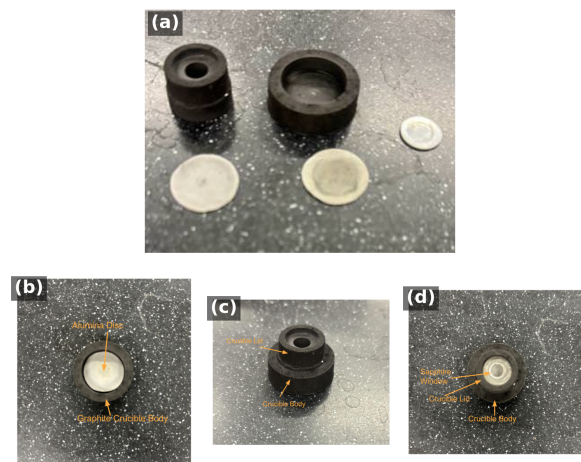


FIG. 2. The two-piece graphite crucible/susceptor used for the nickel anneals reported in this work. (a) Fully disassembled. Top left is the crucible lid, top middle is the bottom of the crucible that acts as the specimen carrier, far right is the sapphire window, bottom left and middle are the alumina discs. (b-d) Specimen-loading sequence. (b) The specimen is placed between two alumina discs in the crucible body. (c) The graphite lid is seated on top. (d) The sapphire window closes the lid bore, leaving an optical path for the pyrometer.

flange. Its upper end mates to the pyrometer housing, a custom 3D-printed part that terminates in a KF40-sized fitting. At that joint the chamber is sealed not by the printed part but by a 55 mm quartz optical disc pressed against an O-ring, and the pyrometer sights through this disc down the chamber axis. Every flanged joint compresses an elastomer O-ring carried on a centering ring, so the vacuum hardware assembles with hand-tightened clamps and no tools. These seals support the 10^{-6} to 10^{-8} Torr operating vacuum. From the bottom of the vacuum column, a tee carries a 0.5 psi overpressure relief valve that protects the quartz tube during backfilling. A flexible bellows continues through the wide-range vacuum gauge to the turbopumping station. The roughing pump sits on a separate support, mechanically decoupled so its vibration does not reach the chamber or the pyrometer sight line. Photographs of the pumping station as installed, together with the relief valve at the base of the vacuum column, are given in the supplementary material. On the gas path, the argon regulator feeds the automated, electrically triggered vent valve. This is a specialized commercial component that admits gas in a controlled way rather than dumping atmospheric pressure into the chamber, which could damage the spinning turbopump. A mass flow controller between them is helpful when continuous inert gas flow is required. On the control path, electrical isolation is maintained between the DAQ wiring and the generator interface. Manual control is verified before automated ramping and soaking are enabled.

A run begins with the specimen being loaded into the crucible (Fig. 2(b-d)), the sample assembly lowered into the quartz tube, and the chamber pumped down. The LabVIEW profile ramps open-loop below the pyrometer floor, switches to closed-loop PID, and runs the programmed soak. Afterwards the sample cools under vacuum, and the chamber is vented only once the turbopump has stopped.

IV. PERFORMANCE VALIDATION

The furnace has completed more than one hundred logged anneals of Ni200 (the standard commercially pure wrought nickel grade) and Ni (4N5, i.e., 99.995 % purity) nickel charges at 900 to 1400 °C, with soaks from minutes to 40 h. The supplementary material provides links to raw run logs for specific samples.

Four steady-state anneals spanning 1200 to 1400 °C were run with one fixed configuration of Ni (4N5) specimen, quartz tube, graphite crucible, and optics. Across these anneals, the soak-mean pyrometer temperature follows the soak-mean power command, expressed in volts on the DAQ's 0–5 V output scale, monotonically and approximately linearly. The relation is well described by $T = 931 + 632 V_P$, with T in °C and V_P the power command in volts (equivalently 9 to 15 % of full scale), and $R^2 = 0.991$, i.e., a high linear response (Fig. S4 of the supplementary material). Its coefficients shift with the charge, crucible, and optical configuration, so any modification to the system calls for recalibration.

A Ni (4N5) anneal at a nominal 1300 °C and 12 h condition serves as the representative closed-loop run. The full annealing curve, including the ramp to temperature, is specified by the user in the control software in advance. The recorded trace is given in Fig. S5 of the supplementary material. The soak held 1302.1 ± 3.0 °C (std. dev.) over the 12 h plateau.

Eight Ni (4N5) anneals at a nominal 1200 °C and 12 h condition were run under the same gas-flow configuration. They reproduced their soak temperature to 1201.2 ± 1.3 °C, a coefficient of variation of 0.11 % (Fig. S6 of the supplementary material). Long-duration stability was demonstrated with two Ni200 anneals at 1325 °C. The first held for 20 h at 1326.5 ± 1.5 °C, and the second held for 40 h. Both traces are shown in Fig. S7 of the supplementary material. Reported temperatures are specific to our setup. Slight variations to the optical pathway or other conditions can affect the temperature readings. By extension, absolute specimen temperature still depends on emissivity and surface condition^{16,21,22}.

V. MICROSTRUCTURAL VALIDATION

The furnace exists to change microstructure, so the decisive test is what the anneals do to real specimens. A

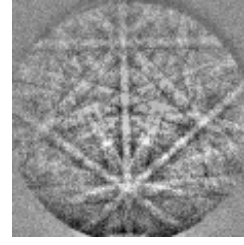


FIG. 3. Raw EBSD Kikuchi pattern from a furnace-annealed nickel specimen with no sample preparation (grinding, polishing, etching, etc.). The specimen was pulled directly from the furnace chamber and placed directly into the EBSD chamber. The pattern is reproduced exactly as saved by the EBSD detector during orientation mapping (the detector's 8×8 -binned output). Even so, the crisply resolved bands and zone axes index directly to the nickel lattice.

striking feature of that test is that it required no metallographic sample preparation at all. The EBSD specimens shown here were never ground, polished, or etched. Samples were able to be pulled from the furnace chamber and placed directly into the EBSD chamber with no processing (e.g., polishing), and each still returned high-quality diffraction patterns rivaling that of high-quality polishing procedures and without scratch artifacts. Figure 3 shows such a high-quality raw EBSD pattern, also called a Kikuchi pattern, from an unprepared furnace-annealed nickel specimen, reproduced exactly as the EBSD detector saved it during orientation mapping. Even at the coarse 8×8 binning used for mapping, the bands and zone axes are crisply resolved. We surmise that preparation-free EBSD on these nickel specimens works here for two reasons. EBSD diffracts from only the top tens of nanometers of the surface, and the long near-solidus anneal removes the deformed near-surface layer that polishing normally exists to remove²³.

In addition, the evacuated, argon-flowing, graphite-enclosed environment holds the oxygen potential low, which plausibly keeps the surface free of an EBSD-obscuring nickel oxide scale²⁴. Rapid nickel surface diffusion at these temperatures can further smooth the surface^{25,26}. The same surface transport produces the deep boundary grooves described below, which can locally reduce indexing right at the boundaries²⁷. Inverse-pole-figure maps built from such patterns resolve hundreds of indexed grains per specimen (Fig. 4). Each indexed color in the map is a crystallographic orientation, so a single map records both the grain sizes and the texture of the annealed specimen. Thermal grooving is itself a record of the anneal. Figure 5 shows edge-on views of a Ni (4N5) specimen annealed at 1200 °C for 12 h and cleaned only ultrasonically. The grooves have deepened enough that they delineate the grain structure on their own, with no etching. Several grain boundaries traverse the full sheet thickness.

Figure 6 shows the microstructure such an anneal produces for a Ni (4N5) specimen soaked 20 h at 1300 °C

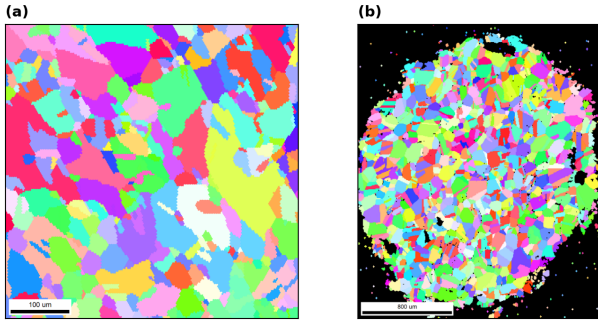


FIG. 4. EBSD inverse-pole-figure (IPF) orientation maps of two furnace-annealed Ni (4N5, 99.995 % purity) specimens, rendered from the recorded EBSD scans. Neither specimen received any sample preparation. (a) A furnace-annealed specimen (100 μm scale bar). (b) A second furnace-annealed specimen (800 μm scale bar). This second specimen was masked for a downstream application related to hydrogen diffusivity measurements, which is why indexed points fill only the circular unmasked region against the black surroundings. The maps resolve hundreds of grains for grain-size and texture analysis. Both specimens appear in the specimen-thermal-history linkage table in the supplementary material.

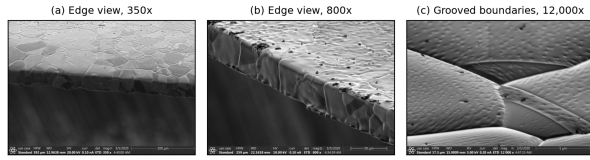


FIG. 5. Grain-boundary thermal grooving on a Ni (4N5) specimen after a 1200 $^{\circ}\text{C}$ / 12 h anneal, imaged edge-on in the as-annealed condition after only an ultrasonic ethanol clean. The grooves have deepened enough that they delineate the grain structure on their own. (a) Edge view at 350 $\times$ (200 μm scale bar). Several grain boundaries traverse the full sheet thickness. (b) Edge and top surface at 800 $\times$ (50 μm scale bar). The grooves outline the surface grain structure. (c) Grooved boundaries at 12,000 $\times$ (5 μm scale bar). Multiple boundaries are grooved, and grooving is visible at several triple junctions. The trenches deepen as grain-boundary energy equilibrates against surface energy at the soak temperature.

under the constant 20 SCCM argon flow (preceded by an evacuation of the chamber to high vacuum and a backfilling purge), with a measured soak of approximately 1301 $^{\circ}\text{C}$. The grains are equiaxed, and thermal grooving allows for the full grain boundary network to be observed under an optical microscope.

A. Extension to ceramic charges: YSZ

The same induction furnace can also be used to process refractory ceramics by modifying only the materials directly in contact with the specimen (the sample assembly) while leaving the generator, vacuum system, pyrometer, cooling system, and control software unchanged.

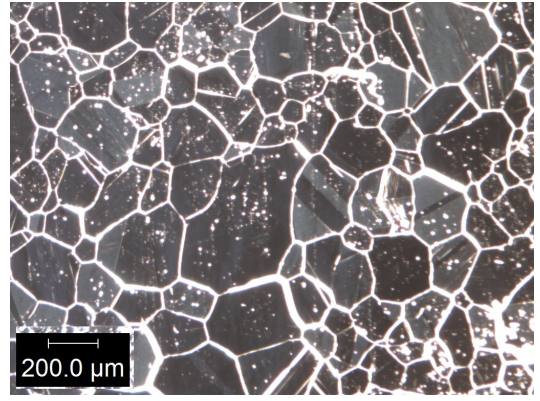


FIG. 6. Optical grain map of a Ni (4N5) specimen after a 1300 $^{\circ}\text{C}$ / 20 h anneal undergoing a constant 20 SCCM argon flow (200 μm scale bar), preceded by an evacuation of the chamber to high vacuum and a backfilling purge directly from high vacuum with argon. The equiaxed grain network is delineated by thermal grooving alone.

Materials that do not couple directly to the RF field require a susceptor. At the temperatures required for grains to coarsen on refractory ceramics, the primary consideration in selecting a susceptor is its chemical compatibility with the specimen and surrounding support materials rather than its RF heating performance. Graphite and tantalum both serve effectively as susceptors, with the preferred choice depending on the operating temperature and the materials in contact with the susceptor.

Grain growth in refractory ceramics is thermally activated. The rate constant of the grain-growth law follows an Arrhenius temperature dependence, so grain coarsening is highly sensitive to annealing temperature^{28,29}. For the 8 mol% YSZ specimens investigated here, grain growth from approximately 10 μm to 80 μm required over 228 h at 1600 $^{\circ}\text{C}$ in a conventional box furnace, whereas the induction furnace coarsened grains from approximately 20 μm to 90 μm in only 45 min at 2500 $^{\circ}\text{C}$ (Fig. 7).

Reaching this temperature range required only minor modifications to the sample assembly (Fig. 8). Direct contact between graphite and alumina was eliminated because alumina is not inert to carbon at these temperatures. Under vacuum or low oxygen potential, carbon carbothermally reduces alumina at the interface. This evolves carbon monoxide and forms aluminum carbides and oxycarbides that degrade the interface^{30–32}. The contact was eliminated by introducing a boron nitride diffusion barrier between the two materials. Alternatively, replacing the graphite with tantalum avoids this compatibility issue while still providing efficient RF coupling.

VI. CONCLUSIONS

A bare commercial RF induction generator can be converted into a computer-controlled, vacuum and gas integrated, pyrometer-feedback annealing system with a



FIG. 7. As-recorded optical micrograph of a YSZ specimen after the 2500 °C / 45 min tantalum-susceptor anneal, which coarsened the grains from approximately 20 to 90 µm. The red traces are the microscope software’s multi-grain size measurements (100 µm scale bar).

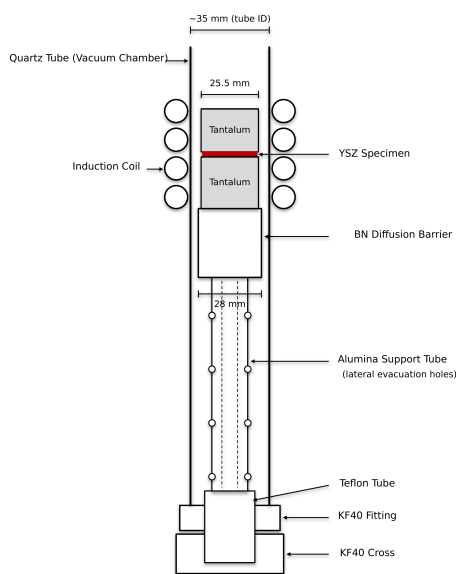


FIG. 8. Sample-assembly configuration for the high-temperature ceramic anneals. A YSZ specimen is sandwiched between two 25.5 mm tantalum susceptor blocks inside the quartz tube. The blocks rest on a boron nitride stub that acts as a diffusion barrier. It prevents reaction between the tantalum and the alumina support tube below, which raises the sample assembly to coil height. The alumina tube has holes bored laterally along its outer perimeter to aid evacuation, and its lower end rests on top of a Teflon tube at the KF40 fitting. The coil, vacuum, pyrometer, and control paths are unchanged from the metal configuration.

documented retrofit. The resulting system is modular, open, and customizable. The vacuum hardware assembles with hand-tightened clamps, the sample assembly is interchangeable, and the LabVIEW control software and design files are provided. Because the retrofit requires only a monotonic analog power-control input, the design transfers across generators and is directly reusable at a small fraction of turn-key cost.

The full system and workflow deliver a high-purity an-

nealing environment. The strongest evidence is that, surprisingly, the annealed nickel specimens needed no metallographic preparation at all. Specimens were annealed, removed from the furnace, and placed directly into the scanning electron microscope, and they returned high-quality EBSD patterns and orientation maps with no grinding, polishing, or etching.

Complete annealing workflows were demonstrated on both nickel and YSZ. On a modern solid-state generator, the nickel campaign produced a power-temperature calibration with $R^2 = 0.991$, soak temperatures reproducible to 0.11 %, stable closed-loop soaks up to 40 h, and the controlled grain growth the system was built to produce. The YSZ campaign highlighted the challenges of material compatibility at extreme temperature. Every failure mode was a contact reaction among the charge, the susceptor, and the support materials, and the interchangeable sample assembly lets a user iterate on those material choices run by run.

High-temperature grain growth was achieved for YSZ in under an hour at 2500 °C in this system, whereas comparable coarsening required 228 h in a standard 1600 °C box furnace. The system is also relatively low-cost compared with commercial systems that reach 2500 °C. In the annealed nickel, many grains and their boundaries span the full sheet thickness, and the deep thermal grooves that mark those boundaries point to columnar grain-growth studies as a natural next application³³.

SUPPLEMENTARY MATERIAL

The supplementary material is organized in seven sections. Sec. SI (hardware) gives the measured crucible part dimensions, the work-coil drawing, and photographs of the vacuum and gas-handling hardware. Sec. SII (thermal performance) collects the power-temperature calibration and the representative-soak, repeatability, and long-soak traces. Sec. SIII (microstructure) presents the raw Kikuchi-pattern survey. Sec. SIV covers the high-temperature extension, with the tantalum-susceptor heat curves and the YSZ micrographs. Sec. SV holds the specimen-thermal-history linkage table, with links to raw run logs. Sec. SVI is the itemized bill of materials, and Sec. SVII is the design-file inventory.

ACKNOWLEDGMENTS

The authors thank Kevin Cole for his support of the original retrofit and acknowledge the Johnson group and the Brigham Young University Department of Mechanical Engineering, which funded the capital equipment cost. This work was supported by the National Science Foundation under Grant No. 1610077. The authors acknowledge the BYU Electron Microscopy Facility for providing access to the equipment and expertise that allowed this project to be performed. This work has also been

made possible, in part, by a grant from the Simmons Research Endowment at Brigham Young University.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

Ethics approval not required. This work did not involve human or animal subjects.

Author Contributions

— provisional: **Sterling G. Baird:** Conceptualization, Software, Investigation, Writing – original draft. **Ryan Weber:** Investigation, Methodology, Data curation, Visualization. **Christopher Nyborg:** Investigation, Methodology, Data curation, Visualization. **Ronnie Guymon:** Visualization, Writing – review & editing. **Gage Erickson:** Visualization, Writing – review & editing. **Oliver Johnson:** Supervision, Project administration.

DATA AVAILABILITY

The data that support the findings of this study are openly available in the development repository (<https://github.com/vertical-cloud-lab/custom-induction-furnace>); an archival snapshot is deposited at Zenodo, <https://doi.org/10.5281/zenodo.20878017>.

- ¹Y.-X. Liu, Z.-J. Ke, R.-H. Li, J.-Q. Song, and J.-J. Ruan, “Study of grain growth in a ni-based superalloy by experiments and cellular automaton model,” *Materials* **14**, 6922 (2021).
- ²H. Mohrbacher, “Application of microalloying for controlling recrystallization and grain growth during downstream steel processing,” *Metallurgical and Materials Transactions A* **57**, 2548–2571 (2026).
- ³K. Alogab, D. Matlock, J. Speer, and H. Kleebe, “The influence of niobium microalloying on austenite grain coarsening behavior of ti-modified sae 8620 steel,” *Isij International* **47**, 307–316 (2007).
- ⁴G. Tian, C. Jia, J. Liu, and B. Hu, “Experimental and simulation on the grain growth of p/m nickel-base superalloy during the heat treatment process,” *Materials & Design* **30**, 433–439 (2009).
- ⁵I. Moore, J. Taylor, M. Tracy, M. Burke, and E. Palmiere, “Grain coarsening behaviour of solution annealed alloy 625 between 600–800 °c,” *Materials Science and Engineering A-structural Materials Properties Microstructure and Processing* **682**, 402–409 (2017).
- ⁶A. F. Zhornyak and V. E. Olikier, “Reduction and decarburization during the annealing of atomized iron-carbon alloy powders,” *Soviet Powder Metallurgy and Metal Ceramics* **21**, 84–93 (1982).

- ⁷Y. Muramatsu, S. Wanikawa, M. Ohtaguchi, H. Okada, and F. Abe, “Gas contamination due to milling atmospheres of mechanical alloying and its effect on impact strength,” *Materials Transactions* **46**, 681–686 (2005).
- ⁸V. Rudnev, D. Loveless, R. L. Cook, and M. Black, *Handbook of Induction Heating*, 1st ed., Manufacturing Engineering and Materials Processing (Marcel Dekker, 2002).
- ⁹O. Lucia, P. Maussion, E. J. Dede, and J. M. Burdío, “Induction heating technology and its applications: Past developments, current technology, and future challenges,” *IEEE Transactions on Industrial Electronics* **61**, 2509–2520 (2014).
- ¹⁰W. Vankan, N. van Hoorn, A. de Wit, and T. Koenis, “Electromagnetic induction heating of TP-CFRP laminates: FEM modelling and validation,” in *Proceedings of the 34th Congress of the International Council of the Aeronautical Sciences (ICAS)* (The International Council of the Aeronautical Sciences (ICAS), 2024).
- ¹¹Y. Xu, A. Leonhardt, P. Birnbaum, J. Gruner, A. Kunke, and T. Clausmeyer, “Design and optimization of induction coil for improved temperature uniformity in fast heating of sheet metal,” *Discover Mechanical Engineering* **5** (2026), 10.1007/s44245-026-00228-5.
- ¹²K. KANDA and H. HASHIMOTO, “Development of high-temperature and low-oxygen atmosphere controlled furnace and its application to metal jointing technology,” *Journal of Advanced Mechanical Design Systems and Manufacturing* **1**, 763–774 (2007).
- ¹³M. Inaba, Y. Honma, T. Hatanaka, and Y. Otake, “Effects of the annealing conditions on the oxidation behavior of fe-36ni alloys,” *Applied Surface Science* **27**, 164–179 (1986).
- ¹⁴A. Tapetado, J. Diaz-Alvarez, M. H. Miguelez, and C. Vazquez, “Two-color pyrometer for process temperature measurement during machining,” *Journal of Lightwave Technology* **34**, 1380–1386 (2016).
- ¹⁵R. Belikov, D. Merges, D. Varentsov, Z. Major, P. Neumayer, P. Hesselbach, M. Schanz, and B. Winkler, “Fast multi-wavelength pyrometer for dynamic temperature measurements,” *International Journal of Thermophysics* **45**, 1–12 (2024).
- ¹⁶S. H. Ham, M. Ferte, G. Fricout, L. Depalo, and C. Carteret, “In-situ spectral emissivity measurement of alloy steels during annealing in controlled atmosphere,” *Proceedings of the 2016 International Conference on Quantitative InfraRed Thermography (QIRT 2016)* (2016), 10.21611/qirt.2016.041.
- ¹⁷F. K. Suleiman, *Improving Pyrometry of Advanced High Strength Steels During Inter-critical Annealing*, Ph.D. thesis, Department of Mechanical and Mechatronics Engineering, University of Waterloo, Waterloo, Ontario, Canada (2025).
- ¹⁸R. Usamentiaga, P. Venegas, J. Guerediaga, L. Vega, J. Molleda, and F. Bulnes, “Infrared thermography for temperature measurement and non-destructive testing,” *Sensors* **14**, 12305–12348 (2014).
- ¹⁹J. Meng, H. Gao, M. Ruan, H. Guo, X. Zhou, and D. Zhang, “Design of vacuum annealing furnace temperature control system based on ga-fuzzy-pid algorithm,” *PLOS ONE* **18**, e0293823 (2023).
- ²⁰E. Rapoport and Y. Pleshivtseva, *Optimal Control of Induction Heating Processes*, 1st ed. (CRC Press, 2007).
- ²¹N. Giuliotti, G. Cosoli, R. Napolitano, G. Pandarese, G. M. Revel, and P. Chiariotti, “Spectral emissivity measurement for high-temperature applications: a systematic review,” *Acta IMEKO* **14**, 1–17 (2025).
- ²²P. Kieruj, D. Przestacki, and T. Chwalczuk, “Determination of emissivity coefficient of heat-resistant super alloys and cemented carbide,” *Archives of Mechanical Technology and Materials* **36**, 30–34 (2016).
- ²³J. I. Goldstein, D. E. Newbury, J. R. Michael, N. W. M. Ritchie, J. H. J. Scott, and D. C. Joy, “Characterizing crystalline materials in the SEM,” in *Scanning Electron Microscopy and X-Ray Microanalysis* (Springer New York, 2018) pp. 491–515.

- ²⁴M. Wolf, "Thermodynamic assessment of the stability of bulk and nanoparticulate cobalt and nickel during dry and steam reforming of methane," *RSC Advances* **11**, 18187–18197 (2021).
- ²⁵P. S. Maiya and J. M. Blakely, "Surface self-diffusion and surface energy of nickel," *Journal of Applied Physics* **38**, 698–704 (1967).
- ²⁶C. V. Thompson, "Solid-state dewetting of thin films," *Annual Review of Materials Research* **42**, 399–434 (2012).
- ²⁷T. A. Gladstone, J. C. Moore, A. J. Wilkinson, and C. R. M. Grovenor, "Grain boundary misorientation and thermal grooving in cube-textured Ni and Ni–Cr tape," *IEEE Transactions on Applied Superconductivity* **11**, 2923–2926 (2001).
- ²⁸S. Tekeli and U. Demir, "Colloidal processing, sintering and static grain growth behaviour of alumina-doped cubic zirconia," *Ceramics International* **31**, 973–980 (2005).
- ²⁹K. Matsui, H. Yoshida, and Y. Ikuhara, "Phase-transformation and grain-growth kinetics in yttria-stabilized tetragonal zirconia polycrystal doped with a small amount of alumina," *Journal of the European Ceramic Society* **30**, 1679–1690 (2010).
- ³⁰J. H. Cox and L. M. Pidgeon, "An investigation of the aluminum–oxygen–carbon system," *Canadian Journal of Chemistry* **41**, 671–683 (1963).
- ³¹M. Kruesi, M. E. Galvez, M. Halmann, and A. Steinfeld, "Solar aluminum production by vacuum carbothermal reduction of alumina—thermodynamic and experimental analyses," *Metallurgical and Materials Transactions B* **42**, 254–260 (2011).
- ³²M. S. Walker and R. J. Fruehan, "Experimental verification of the thermodynamics of critical aspects of the carbothermic reduction of alumina. Part 1: Experimental result of reacting Al_2O_3 –C mixtures," *Metallurgical and Materials Transactions B* **44**, 853–861 (2013).
- ³³Z. Zhang, G. Chen, and G. Chen, "Dynamics and mechanism of columnar grain growth of pure iron under directional annealing," *Acta Materialia* **55**, 5988–5998 (2007).